

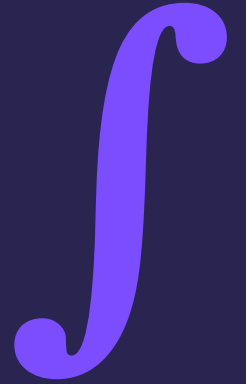
IB MATH AA SL

Lesson 42

Definite Integrals & Area

Calculus · 60-minute lesson

Syllabus SL 5.5 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. Find $\int 2x \, dx$.
2. Evaluate x^2 at $x = 3$ then at $x = 1$.
3. $3^2 - 1^2 = ?$

Answers (click to reveal)

1. $x^2 + C$
2. 9 and 1
3. 8

What we're learning today

LEARNING OBJECTIVES

- 1 Evaluate**
a definite integral.
- 2 Apply**
the limits (top minus bottom).
- 3 Find**
the area under a curve.
- 4 Use**
the GDC for definite integrals.

SUCCESS CRITERIA — I CAN...

- ✓
integrate, then substitute the limits
- ✓
compute $F(b) - F(a)$
- ✓
interpret a definite integral as area
- ✓
use the GDC to check the value

Definite integrals

WITH LIMITS

$$\int_a^b f(x) dx = F(b) - F(a)$$

Integrate

find $F(x)$ (no + C needed here)

Substitute

top limit minus bottom limit

NO + C

The constant cancels in $F(b) - F(a)$.

So

omit C for a definite integral.

Evaluate a definite integral

QUESTION

Evaluate $\int_1^3 2x \, dx$.

SOLUTION

Integrate: $F(x) = x^2$.

$$\int_1^3 2x \, dx = [x^2]_1^3$$

Top minus bottom ($3^2 - 1^2$):

$$= 9 - 1 = 8$$

Your turn — definite

ANSWER IN YOUR BOOK

1. $\int_0^2 2x \, dx$.

2. $\int_1^2 3x^2 \, dx$.

3. $\int_0^1 4x \, dx$.

Answers (click to reveal)

1. $[x^2]_0^2 = 4$

2. $[x^3]_1^2 = 8 - 1 = 7$

3. $[2x^2]_0^1 = 2$

Area under a curve

AREA = INTEGRAL

$$\text{Area} = \int_a^b f(x) \, dx$$

Above the x-axis

the definite integral gives the area

GDC

use the definite-integral function to evaluate

CAREFUL

Below the axis

gives a negative — area is its absolute value.

Limits

a and b are the x-bounds of the region.

Area under a curve

QUESTION

Find the area between $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 2$.

SOLUTION

Area = $\int_0^2 x^2 dx$; integrate to $x^3/3$:

$$\left[\frac{x^3}{3} \right]_0^2$$

Top minus bottom ($8/3 - 0$):

$$= \frac{8}{3}$$

Exam-style question

(a) Evaluate $\int_1^4 (2x + 1) \, dx$. [3]

(b) Hence state the area between $y = 2x + 1$, the x-axis and $x = 1$, $x = 4$. [1]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $[x^2 + x]_1^4 = (16+4) - (1+1)$ (M1)(A1) = 18 (A1)

(b) area = 18 (square units) (A1)

Common mistakes



No + C

Drop the constant for a definite integral.



Order

It is $F(b) - F(a)$: top limit first.



Negative area

A region below the x-axis integrates to a negative.



Square brackets

Write $[F(x)]_a^b$ to show the limits.

Mixed practice

QUESTIONS

1. $\int_0^3 2x \, dx$.

2. $\int_1^2 4x \, dx$.

3. Area under $y = x^2$ from 0 to 3.

4. $\int_0^1 6x^2 \, dx$.

Answers (click to reveal)

1. $[x^2]_0^3 = 9$

2. $[2x^2]_1^2 = 8 - 2 = 6$

3. $[x^3/3]_0^3 = 9$

4. $[2x^3]_0^1 = 2$

Plenary & exit ticket

KEY FORMULAS

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

$$\text{Area} = \int_a^b f(x) \, dx$$

EXIT TICKET — before you leave

1. $\int_0^2 2x \, dx$.

2. Do you add + C for a definite integral?

3. $\int_1^2 2x \, dx$?

Answers (click to reveal)

1. 4

2. no

3. $[x^2]_1^2 = 3$

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Definite integrals & area.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 43

End of Topic Test — Calculus. Revise Lessons 34–42.